



A page from Avicenna's *Canon of Medicine* shows the heart and skull as part of an illustration explaining the theory of the four humors.

“ NOW IT IS **ESTABLISHED** IN **THE SCIENCES** THAT NO **KNOWLEDGE** IS ACQUIRED SAVE THROUGH THE **STUDY** OF ITS **CAUSES** AND **BEGINNINGS**. ”

Ibn Sina (Avicenna), Arab polymath, from *Canon of Medicine*, c.1005

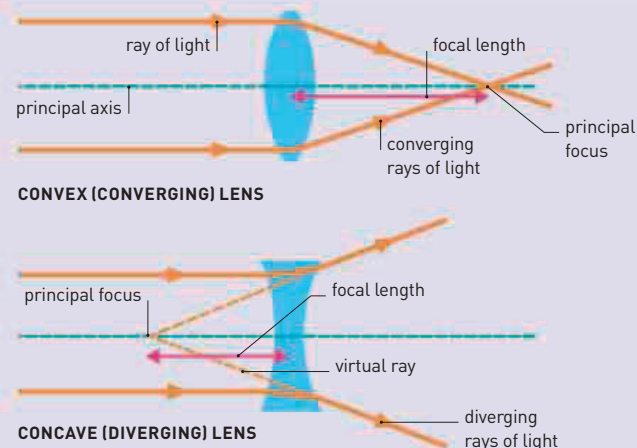
AROUND 1005, THE ARAB MEDICAL SCHOLAR and polymath **Ibn Sina** (known as Avicenna in Europe) wrote the *Canon of Medicine*, a major compendium that sought to provide a systematic understanding of medical knowledge of the time. Avicenna tried to **reconcile theories** of four humors (blood, yellow bile, black bile, and phlegm; see 100–250) with **Aristotle's idea** of three life forces (psychic, natural, and human). Avicenna's careful and comprehensive account in **five volumes** of physiology, diagnosis, therapy, the pathology of diseases, and pharmacology made it an **extremely valuable medical handbook**. It was commented on by many subsequent Arabic physicians and was printed as Latin translations 36 times.

The Persian astronomer and mathematician **Abu Sahl al-Quhi** (c.940–1000) was **head of the observatory** founded by Sharaf al-Dawla in Baghdad in 988, but he was particularly noted for his work in **solving equations of greater than the second degree** (in which the highest power of a variable is more than two). He employed a geometrical method of intersecting curved lines to achieve this. Around 1000, he wrote *On the Construction of an Equilateral Pentagon in a Known Square*, in which he demonstrated the solution by solving an equation to the fourth degree.

In 1005, the Fatimid caliph **al-Hakim** founded the **House of Knowledge (Dar al-'ilm)** in Cairo. Equipped with a vast library covering subjects ranging

HOW LENSES WORK

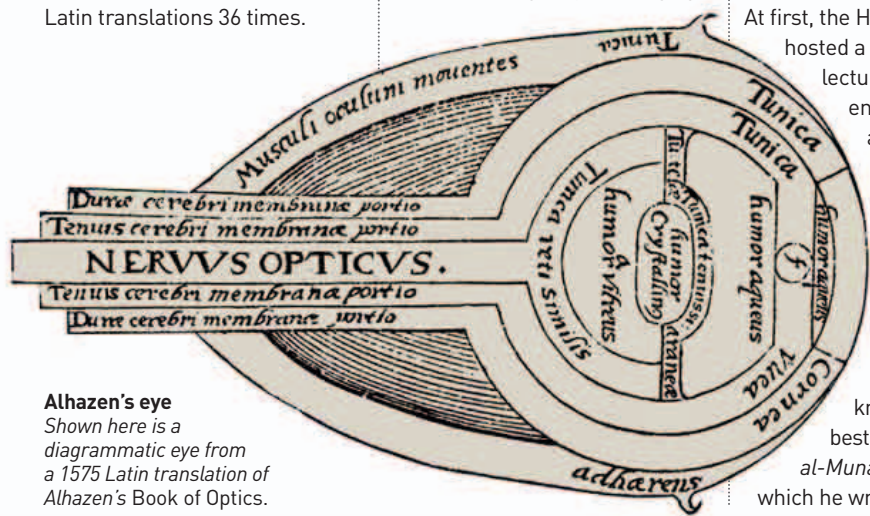
A convex lens is thicker in the middle than at the sides. When light rays strike the lens they are diffracted and converge behind the lens at a single point called the principal focus. Convex lenses are used to treat far-sightedness because they bring objects that are closer, into focus. A concave lens is thinner in the middle than at the sides. Light rays diverge and seem to focus in front of the lens. These lenses address near-sightedness.



from Islamic philosophy and law to physics and astronomy, it became a **center for philosophers and theologians**. At first, the House of Knowledge hosted a series of public lectures, but these ended in 1015 after fears that religious dissidents were establishing a presence there. The Arabic sage **Abu 'Ali ibn al-Haytham** (c.965–1039), also known as **Alhazen**, is best known for his *Kitab al-Munazir* (*Book of Optics*), which he wrote between 1011

and 1021. He proposed that the blinding effect of bright light and the existence of after-images proved that **vision was caused by light coming into the eye**. He also developed a new theory of

the eye's physiology, describing it as being made up of various **humors** and separated into sections by **spherical sheaths**.



Alhazen's eye
Shown here is a diagrammatic eye from a 1575 Latin translation of Alhazen's *Book of Optics*.



IBN SINA (980–1037)

Born near Bukhara, Uzbekistan, Ibn Sina (Avicenna) was a medical prodigy. He claimed to have successfully treated patients by age 16. He served the Samanid rulers of Bukhara, but their overthrow in 999 led to his exile. He ended up at the court of Shams al-Dawla of Hamadan, where he wrote his great *Canon of Medicine*.

1000 Al-Quhi describes the solution of equations beyond the second degree

1004 Caliph al-Hakim founds the **House of Knowledge** in Cairo

1006 Muslim astronomer Ali ibn Ridwan gives first description of a **supernova**

1005 Ibn Sina (Avicenna) composes the *Canon of Medicine*, a **medical compendium**

1011–21 Alhazen proposes his theory of vision in *Book of Optics*

1015 Arab physician Masawaih al-Marindi prescribes the **gem electuary**—a paste made of ground gems—to cure melancholia